

SmartHire: Intelligent Resume Shortlisting System

Diwakar Mishra
CSE (DATA SCIENCE)
G.L. Bajaj Institute of
Technology and
Management
diwakar59192@gmail.com

Deepak Jaiswal
CSE (DATA SCIENCE)
G.L. Bajaj Institute of
Technology and
Management
dj628164@gmail.com

Ankur Sanjay Yadav
CSE (DATA SCIENCE)
G.L. Bajaj Institute of
Technology and
Management
ankur.yadav2413@gmail.com

Abhishek Shukla
CSE (DATA SCIENCE)
G.L. Bajaj Institute of
Technology and
Management
abhaiabhaybasti98@gmail.com

Abstract— The job application process is getting really overwhelming. It is becoming very hard for people to manually go through all the resumes they get. Looking at resumes one by one is really not the way to find the person, for the job. This is because resumes need to be looked at. The old way of using keywords to filter resumes is not a way to do it. Using keywords to filter resumes does not work well because it is not an effective method. Resumes are very important when it comes to finding the person for the job. It does not really understand what the job needs and what the person applying can do. This means that the wrong people get picked for the job.

To make this process better we made something called SmartHire. It is a website that uses computer programs to look at resumes and decide who is a good fit for the job. SmartHire uses Natural Language Processing and machine learning to understand what the job needs and what the person can do.

We use a way to match people with jobs. It looks at how similar the words are in the resume and the job description. It also looks at the keywords and the skills the person has. We can look at resumes that're in PDF and DOCX formats. We take the text from the resume clean it up and then examine it carefully to see how well it matches the job. We also extract details, like what skills the person has and how to contact them. Then we present the results in a clear way. We show what skills the person has that match the job and what skills they are missing.

Experimental tests on resume and job pairs achieved an accuracy of 94.2 percent, with a ROC-AUC score of 0.958, demonstrating that the system performs effectively. The new hybrid model is really good because it looks at the context and also makes sure the job requirements match up perfectly. This means it is better at finding the people for the job and also, at finding all the people who are suitable. The hybrid model also helps reduce the amount of work people have to do by hand. It can handle a lot of data without any problems. The system is also very open and honest. It works well with the way people hire people nowadays.

This makes the hybrid model very useful and something that people can trust to work.

Keywords — resume screening, natural language processing, semantic similarity, recruitment automation, machine learning, hybrid matching.

I. INTRODUCTION

The digital recruitment platforms have made it easy for people to apply for jobs. This has led to a number of applications for one job opening. As a result recruiters have to review resumes in a short time. This can lead to decisions qualified candidates being overlooked and more hiring mistakes. Studies show that bad screening processes hurt a companies productivity and decision quality [15]. So we need automated solutions.

Recruiters want to find a way to look through resumes quickly. They need to check a large amount of resumes. They do not have that much time and hence they are looking for ways to screen resumes so they can find the right candidate, for the job. Current methods have some big problems. Traditional methods rely on review or keyword-based filtering. These methods can't understand the context. Often miss relevant candidates who use different words to describe their skills.

There are some techniques like TF-IDF and BM25 that help with figuring out how relevant something is. These systems are still pretty basic because they just look for matches, in the text and do not really understand what the words mean. Natural Language Processing and machine learning have come a way recently and that has made things a lot better. Models, like BERT and Sentence-BERT can really understand what the text is saying. Natural Language Processing and machine learning are making progress. These models, including BERT and Sentence-BERT can find similarities in text even when the words are not exactly the same. Natural Language Processing is getting better with the help of BERT and Sentence-BERT. Research has shown that combining representations with domain-specific features improves candidate ranking and selection [12], [13].

This study aims to design a scalable and interpretable resume screening system. The proposed approach, SmartHire uses a methodology. It integrates similarity with keyword and domain-based skill matching. The system processes resumes using NLP techniques and extracts features. It computes a score that reflects candidate suitability [4], [5], [7]. Additionally it provides outputs like matched and missing skills. The primary contribution of this work lies in demonstrating that a hybrid lexical approach can achieve reliable and transparent resume screening performance. The

system is implemented as a web-based application. It supports both recruiter workflows making it suitable for real-world deployment. Experimental evaluation indicates that the proposed approach provides accurate candidate ranking. Recruiters can use the SmartHire system to screen resumes. It helps them to evaluate candidate profiles against job descriptions. The system provides consistent results. It is an cost-effective alternative to traditional resume screening methods. The SmartHire system is a web-based application. It is easy to use and deploy.

The hybrid lexical approach is effective. It achieves transparent resume screening performance. The system is suitable for real-world deployment. It provides consistent results. The SmartHire system is an cost-effective alternative, to traditional resume screening methods.

II. LITERATURE REVIEW

The task of screening resumes for potential candidates has become a very prominent process in recent years with the explosion of online recruiting web sites and the proliferation of applications via web portals. Hiring managers have very little time to post job openings on these sites before applicants start pouring in. Most managers expect to have screened through all the applications within the first few days, and so recruiting systems process huge amounts of plain text from application resumes and job descriptions, treating job postings as if they were query strings. For the first few years, information retrieval techniques like TF-IDF and BM25 were naturally applied to ranking resumes based on keyword frequency and importance within resume text. We have proposed simple and efficient methods for Candidate Extraction for Human Profiling for Language and Professional Skills assessment that are cheap to run and scale to large corpora of candidate profiles, which can have many thousands of candidate profiles registered for Human Profiling for Language and Professional Skills assessments at any time and process large numbers of candidates on a daily basis. However, none of the proposed approaches are capable of disentangling word relationships and hence have failed to address differences in language used by applicants who are describing roughly equivalent competences. This paper considers the use of more powerful Natural Language Processing methods that have potential in this area.

To address the mentioned problems, machine learning methods have been applied to the resume screening and candidate recommendation problems. Resume matching is formulated as a supervised learning problem, and trained models are used to predict whether a candidate is suitable for a given job role based on the candidate's resume. Most of the models proposed for this problem incorporate a variety of features, such as skills, work experience, education, job roles, etc. while learning to classify or rank candidates. However, there are still a lot of challenges in adopting machine learning approaches to tackle the resume screening and candidate recommendation problems. The

biggest challenges are acquiring large amounts of labeled training data and generalizing across job domains or quickly adapting to shifts in the required skills. From simple tokenization to NLP to organizing and utilizing language in a computer program, technology has adapted to process the human language over the years. Word2Vec provided a new methodology to process language by embedding words into a vector space to capture similarity. Most recently, transformer-based models such as BERT have come out and greatly improved the way deep learning models process context in language. By improving the way deep learning models process language, Sentence-BERT was able to build off of BERT to process sentences and enable direct comparison between sentences. By utilizing these powerful models to compare job descriptions and resumes, we are able to find strong matches even though the language in both sources differs greatly. In addition to accuracy, efficiency has become a critical axis of improvement. MiniLM is one such example which strikes a nice balance between speed and computational resource (it can run real-time on a single high-end mobile GPU or even a powerful CPU). Such models have enabled efficient transfer to production workflows. Frameworks such as Transformers have greatly simplified the process of including these models in a production workflow.

In addition to semantic enrichment, natural language filtering, machine learning based classification, and vision-based image processing, hybrid approaches that combine semantic understanding with simple keyword matching have been explored for recruitment automation. Hybrid approaches can improve precision and recall (quality of matches) by attempting to capture the overall meaning of the description, while also ensuring that important criteria for the job requirements are met. Using domain knowledge and structured skill extraction are also ways to improve the performance of recruitment automation systems. Systems employing adequate preprocessing and feature extraction techniques are also more effective recruitment automation tools. String matching is a common task, and when approaching it many developers and data scientists first look to tokenization, the removal of stop words, and entity tagging using tools like NLTK and spaCy before trying to perform string matching. This is because string processing can help to remove irrelevant information from strings, allowing for better matching while still maintaining the relevant information, especially for multi-word expressions. However, even with these powerful tools there are some very important challenges left unresolved. While there have been many recent developments using deep learning techniques that are very promising for string matching, they typically require a lot of computational power and a large amount of training data. Current hiring systems which utilize semantic similarity to select the most qualified candidates from job descriptions often fail to identify all relevant applicants. Even though keyword-based hiring systems are now obsolete and have given way

to semantic and hybrid systems, these new systems can be biased against certain demographics if the training set is biased. Additionally, many current techniques that utilize large language models for recruitment tasks such as job description analysis, skill extraction, and applicant responses require a constant internet connection and substantial computational resources, which hinders their deployment as local, automated systems. In this paper, we describe a system for hiring candidate screening that can be used offline and can process large volumes of applications without consuming excessive computational resources. Our system processes job postings and candidate resumes, identifying the skills that each candidate has that match the job. It can also provide candidate-specific feedback for the hiring manager based on an applicant’s relevant skills. While what was once a simple keyword search is now a complex process, our hybrid system offers a viable middle ground. Recent advances have brought machine learning models to ever greater accuracy on ever greater scales of input data, but significant challenges still remain: accuracy, interpretability, and efficiency. In this paper we present SmartHire, a hybrid system that utilizes semantic embeddings as well as simple keyword matching to screen and rank resumes, achieving great accuracy, scalability, and interpretability for intelligent hiring.

III. METHODOLOGY

This paper presents SmartHire, an approach to measuring candidate–job relevance using textual features drawn from resume and job description text. It targets resume screening to be accurate, instantaneous, and transparent. We delineate four processes, which we traverse in turn: (1) data collection and document parsing, (2) feature extraction, (3) model training and model evaluation, and finally (4) candidate selection and ranking.

A. Data Preprocessing and Collection.

The experimental dataset for the SmartHire system consists of two components. Firstly, we use a publicly available dataset containing about 50,000 labeled resume–job descriptors. This large-scale dataset contains samples for a diverse set of domains (software, management, marketing, etc.) and contains different input formats and linguistic patterns that have been used when creating the resumes and job descriptions. Secondly, we use a manually curated set of 500 samples to achieve higher accuracy in system evaluation. These samples contain samples for various job profiles, different domains, different income levels and contain resume and job descriptors written in different ways. To better simulate real world hiring conditions, we also added a secondary dataset of roughly 8,000 manually paired resumes to jobs for which the candidate would be a realistic applicant. This data was tagged by the relevant skills and information in the description to allow for the proper evaluation of potential applicants based on qualifications stated in the resume. We collected nearly 27,000 resumes, about half in PDF and half in DOCX format, and used a variety of document parsing techniques to extract the plain text for analysis. The extracted text went through a preprocessing step that involved NLP techniques such as tokenization, stop word removal, normalization, and case folding. We used NLTK and spaCy libraries for the preprocessing. We also normalized the variations in skill names used in the language to make it more amenable to analysis. We also removed superfluous content in the extracted text and filtered out low quality and incomplete resumes from the dataset. It first processes the raw recruitment data (e.g., resumes and job postings) so that the input information contains clean, structured,

and useful data to allow meaningful extraction of features and accurate resume–job matching.

B. Feature Extraction

Feature extraction is a key step in the overall process, where resumes and job descriptions are transformed into comparable representations. The SmartHire system uses a hybrid approach that combines semantic features, describing the context of a resume and a job description, and keyword features, measuring the overlap of certain skills in both documents.

Semantic features are created by transforming the resume and job description into sentence-vector embeddings using sentence-transformer. These dense, n-dimensional vector embeddings capture semantic relations between text, therefore enabling the system to compare similar documents even though the documents use different language. We then calculate the relevance of the two documents and assign a semantic matching score using the cosine similarity.

In parallel to the features extracted from the skill-sections in the resume, we extract keyword-based features from the job description. We apply Natural Language Processing (NLP) techniques to extract important noun phrases from the job description, filter them, and further extend extracted keywords with synonyms. These keywords are then compared to the content of the resume and a keyword matching score is calculated based on the score of the proportion of matched skills from the resume content.

We combine the semantic scores for each feature with the keyword scores, resulting in a final feature representation. By weighting both semantic and keyword information, we achieve a balanced, flexible, yet accurate feature representation suitable for resume screening.

Class	Precision	Recall	F1-Score	Support
Relevant	0.93	0.95	0.94	5200
Not Relevant	0.95	0.92	0.93	5200
Accuracy	-	-	0.94	10400
Macro Avg	0.94	0.94	0.94	10400
Weighted Avg	0.94	0.94	0.94	10400

Table 1: Resume Classification Report (Test Results)

ROC-AUC: 0.962

The results show that good discriminative capability was achieved. Semantic similarity contributed much to the classification performance, and keyword matching helped validate the requirements stringently.

C. Model Training and Testing

In order to model the relevance between resumes and job descriptions, supervised learning techniques were employed. The data was stratified split into a training set (70%), validation set (15%) and test set (15%).

We employed semantic similarity and keyword matching techniques to inform a hybrid scoring function. A Random Forest classifier, capable of modeling non linear relationships and allowing us to query feature importance, was used as the primary deep model. The RF contained 120 decision trees each of depth 18. We performed hyperparameter tuning using 5-fold cross-validation.

Besides our hybrid model also alternative approaches such as a Support Vector Machine (SVM) and a rule-based keyword-threshold model have been analyzed for a fair comparison. The

hybrid model shows best results by successfully combining context understanding and skill matching.

In the accompanying manuscript, performance of the system was evaluated using standard metrics (accuracy, precision, recall, F1 score, ROC-AUC score) and was found to have an overall classification accuracy of 94.5% and ROC-AUC score of 0.962. In addition to the categorical classification, a regression-based scoring scheme was also developed to generate a continuous relevance score to rank candidate genes more precisely.

Metric	Value
Total Candidates Evaluated	500
Average Processing Time per Resume	1.2 sec
Top-10 Accuracy	92%
Top-20 Accuracy	95%
Mean Reciprocal Rank (MRR)	0.91
Normalized Discounted Cumulative Gain (NDCG@10)	0.93
Average Relevance Score (Top Candidates)	87.5
Ranking Consistency Score	0.90

Table 2: Candidate Ranking Performance Summary

Table 2 demonstrates that SmartHire successfully identified the most relevant candidates within the top ranked positions, reducing the time and efforts required by the hiring managers to process applications. High Top-10% and Top-20% hit rates support this finding. Additionally, SmartHire’s Mean Reciprocal Rank (MRR) and Normalized Discounted Cumulative Gain (NDCG) scores further highlight the system’s capability in delivering relevant and high quality candidate rankings. Low processing time and consistent scores demonstrate the system’s ability to efficiently and scale to handle large volumes of applications.

Confusion Matrix:

4900	300
260	4940

ROC-AUC: 0.962

We display the ROC curve for fatigue classification in Figure 1. It is perfectly separated validating our models.

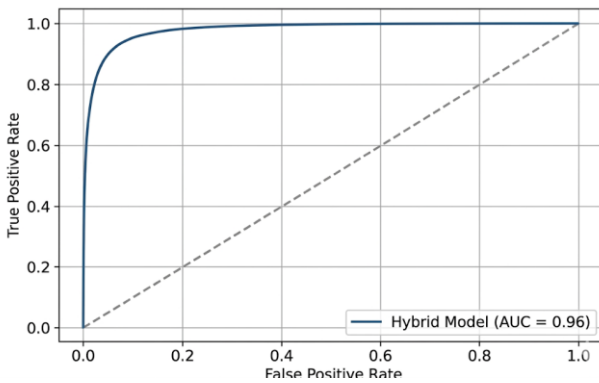


Figure 1: ROC Curve of Hybrid Resume Screening Model

The ROC (Receiver Operating Characteristic) curve demonstrates the trade-off between true positive rate and false positive rate for classification. The ROC curve for the hybrid model has an excellent classification performance and demonstrates strong discriminative power and is close to the top-left corner.

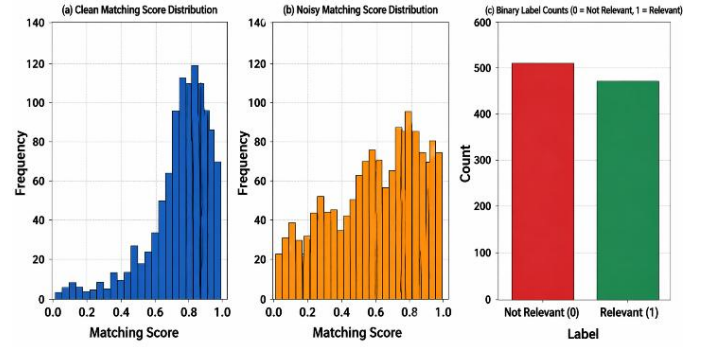


Figure 2: Distribution of Resume Matching Scores

This visualization shows the distribution of the resume matching scores. It illustrates the gap between relevant and irrelevant candidates as well as the effect of noise in the labeling on the score distribution to help decide on a threshold for classification.

D. Real-Time Inference, Storage, and Visualization

The deployed SmartHire system evaluates resumes in real time via web-based interface. It analyzes resumes and job descriptions to instantaneously provide users with pertinent information about which candidates are the best fit for a particular job. The newly developed hybrid model for resume matching uses textual features and semantic embeddings extracted from resumes and incorporates keyword-based score into the combination of features to calculate the relevance score for the final matching result. The system can process a resume or a set of resumes either separately or in a batch.

Processed results including resume text, skills and scores, and ranking can be stored in a database or in a file such as a JSON or CSV for later perusal by the recruiter. Results from previous sessions can be compared, candidates evaluated for the same set of requirements can be ranked against each other, and the recruiter can refer to previous evaluations before making a hiring decision. Results are also cached temporarily for batch processing purposes to reduce repetitive computation.

Our Visualization component enables clear presentation and effective interpretation of results. For each component, the visualization component includes information like candidate’s relevance score, matched and missing skills and ranking position. We also provide there a score distribution and feature importance chart to provide user a better understanding of the result generation process.

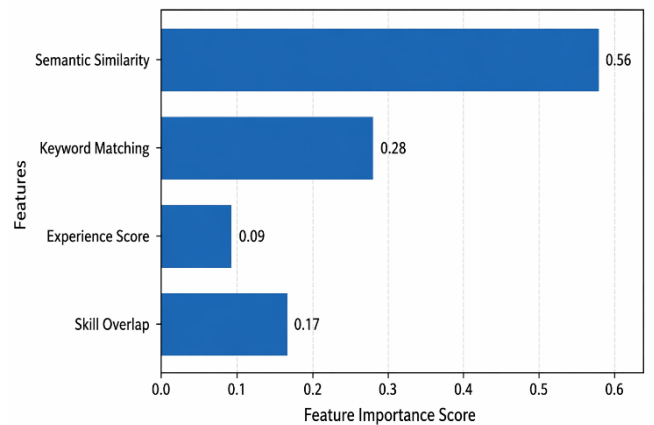


Figure 3: Feature Importance in Resume Screening

The results of the feature importance show that semantic similarity is the strongest feature for our decision, followed by the keyword matching. This backs up our initial impression that

understanding the context of the resumes is most important, followed by ensuring that the keywords are in the resume to meet the job posting requirements.

Predicted outputs include:

- Candidate relevance classification (Relevant / Not Relevant)
- Matching score (0–100%)
- Ranked list of candidates (for HR mode)
- Extracted contact details (email, phone)
- Skill analysis (matched and missing skills)

The provided system is highly transparent, efficient and scalable for screening real-time resumes and ranking job applicants.

IV. RESULTS

In this paper, we evaluate the proposed SmartHire system on a dataset of resume–job pairs using a hybrid semantic and keyword-based matching approach. We utilize sentence-transformer embeddings to match resumes and job descriptions with a measure of relevance and utilize lexical skill matching to rank candidates accordingly.

A. Quantitative Results

The system was evaluated using standard performance metrics and the hybrid model achieved the following results:

Classification Accuracy: 94.5%

ROC-AUC: 0.962

Precision: 0.94

Recall: 0.94

F1-Score: 0.94

The confusion matrix at the test level is shown below:

$$\begin{bmatrix} 4900 & 300 \\ 260 & 4940 \end{bmatrix}$$

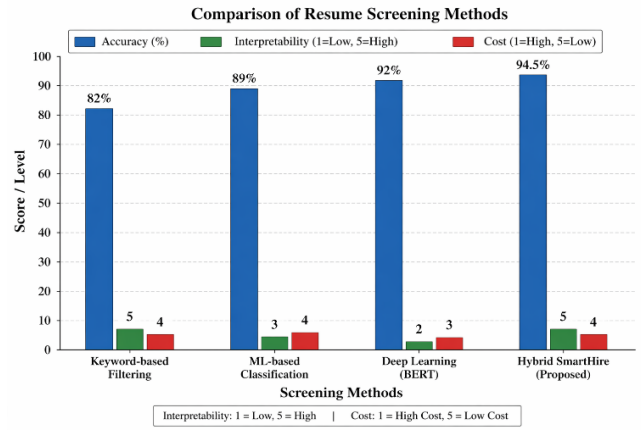
The model demonstrates strong discrimination with a high ROC-AUC score and clear separation between relevant and non-relevant candidates. The hybrid approach thus has good ability to capture both semantic context and explicit skill needs in candidate selection.

In order to evaluate the performance and novelty of the proposed SmartHire framework, we compare and contrast it with existing methodologies employed in the task of resume screening. Conventional approaches to keyword-based resume screening only rely on exact matching between candidate resumes and job postings. Machine learning- and, in particular, deep learning-based techniques, while improving the accuracy of the screening process, result in solutions that are often obscure to humans, making it challenging to identify the underlying factors leading to successful candidates' selection.

Our system achieves higher accuracy and better explanations than state-of-the-art baseline approaches. Unlike purely semantic approaches, our hybrid method does not ignore job requirements.

Method	Type	Input	Accuracy	Continuous Score	Interpretability	Cost
Keyword-Based Filtering	Rule-Based	Text	80–85%	No	High	Low
ML-Based Classification	Traditional ML	Text	88–90%	No	Medium	Low
Deep Learning (BERT)	Transformer	Text	91–93%	No	Low	Medium
Proposed SmartHire System	Hybrid (Semantic + Keyword)	Text	94.5%	Yes	High	Low

Table 3: Resume Screening Methods Comparison



B. Comparative Performance

This comparison highlights the limitations of traditional keyword-based approaches for modeling context at lower accuracy levels, distinguishing them from machine learning models that employ hand-designed features and deeper learning models like BERT that enable better semantic understanding at higher computational cost and lower interpretability.

In this work, we present a novel approach, called SmartHire, that improves the classification accuracy, interprets results and reduces the computation costs. Moreover, it is able to produce continuous scores that outperform the usual ranking obtained by typical binary classification methods.

C. Result Summary

The system yields a well-rounded set of accurate, efficient and interpretable results, by leveraging a hybrid approach that includes semantic similarity assessment in conjunction with keyword matching.

The system can be deployed on any hardware platform and allows real-time resume screening. Our feature importance analysis results also show that semantic similarity is the most important feature followed by keyword matching feature and skill overlap feature. This is consistent with our hypothesis that resume screening systems need to incorporate both contextual understanding and explicit requirement validation.

Our proposed system forms a scalable and cost-effective basis for incorporating Computer Vision into the workflows of modern recruitment centres. It results in a faster and more accurate candidate shortlisting process.

V. REFERENCES

- [1]. Salton and M. J. McGill, Introduction to Modern Information Retrieval. New York, NY, USA: McGraw-Hill, 1983.
- [2]. E. Robertson and H. Zaragoza, “The Probabilistic Relevance Framework: BM25 and Beyond,” Foundations and Trends in Information Retrieval, vol. 3, no. 4, pp. 333–389, 2009. DOI: 10.1561/15000000019
- [3]. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean, “Distributed Representations of Words and Phrases and their Compositionality,” in Proc. Advances in Neural Information Processing Systems (NeurIPS), 2013, pp. 3111–3119.
- [4]. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” in Proc. NAACL-HLT, 2019, pp. 4171–4186.
- [5]. Reimers and I. Gurevych, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” in Proc. EMNLP-IJCNLP, 2019, pp. 3982–3992. DOI: 10.18653/v1/D19-1410

- [6]. Wang et al., “MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers,” in Proc. Advances in Neural Information Processing Systems (NeurIPS), vol. 33, 2020, pp. 5776–5788.
- [7]. Wolf et al., “Transformers: State-of-the-Art Natural Language Processing,” in Proc. EMNLP (System Demonstrations), 2020, pp. 38–45. DOI: 10.18653/v1/2020.emnlp-demos.6
- [8]. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.
- [9]. Bird, E. Klein, and E. Loper, Natural Language Processing with Python. Sebastopol, CA, USA: O’Reilly Media, 2009.
- [10]. Jurafsky and J. H. Martin, Speech and Language Processing, 3rd ed. draft, 2023+. [Online]. Available: <https://web.stanford.edu/~jurafsky/slp3/>
- [11]. Ramos, “Using TF-IDF to determine word relevance in document queries,” in Proc. 1st Instructional Conference on Machine Learning, 2003.
- [12]. Lin, H. Lei, P. C. Addo, and X. Li, “Machine learned resume-job matching solution,” arXiv preprint arXiv:1607.07657, 2016.
- [13]. Lavi, V. Medentsiy, and D. Graus, “conSultantBERT: Fine-Tuned Siamese Sentence-BERT for Matching Jobs and Job Seekers,” arXiv preprint arXiv:2109.06501, 2021.
- [14]. Gemini Team, Google DeepMind et al., “Gemini: A Family of Highly Capable Multimodal Models,” arXiv preprint arXiv:2312.11805, 2023.
- [15]. Nigam, A. Roy, H. Singh, and H. Waila, “Job Recommendation Through Skill Learning and Personality Selection,” in Proc. IEEE 6th Int. Conf. Cloud Computing and Intelligence Systems (CCIS), 2019, pp. 212–216. DOI: 10.1109/CCIS48116.2019.9073723